

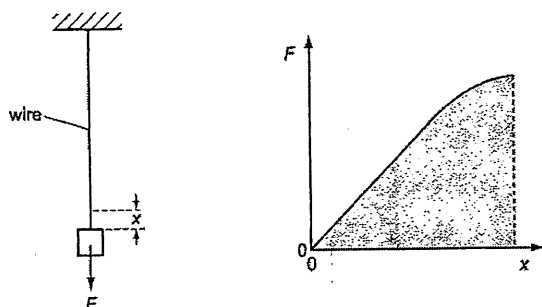


Important note: *denotes questions that may require knowledge from later chapters in order to solve them.

I. MCQs

1. [SAJC/Prelims 2014/P1/9]

A wire, fixed at its upper end, is subjected to an increasing load F by increasing the mass attached to its lower end. A graph of F against the extension x of the wire is shown.

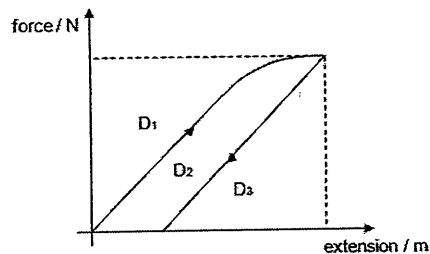


The wire is stretched beyond its elastic limit.
What does the shaded area on the graph represent?

- A the amount of elastic potential energy stored in the wire
- B the amount of heat produced in the wire
- C the loss of gravitational potential energy of the mass
- D the work done by F on the wire

2. [SRJC/Prelims 2014/P1/8]

The stretching of a particular material and the easing of the stretched material take different paths in a force-extension graph as shown.

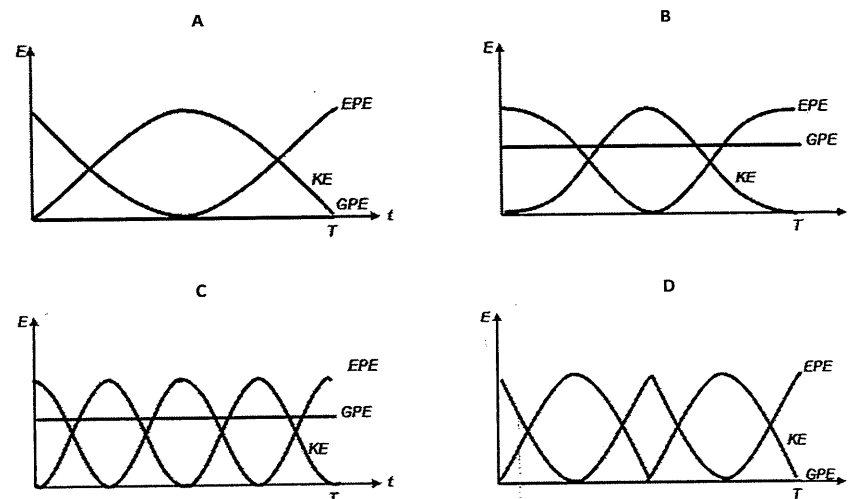


- A $D_1 + D_2$ only
- B D_2 only
- C $D_2 + D_3$ only
- D D_3 only

3. [ACJC/Prelims 2014/P1/11]*

A mass on a smooth horizontal table is attached by two light springs to two fixed supports. The mass executes linear simple harmonic motion.

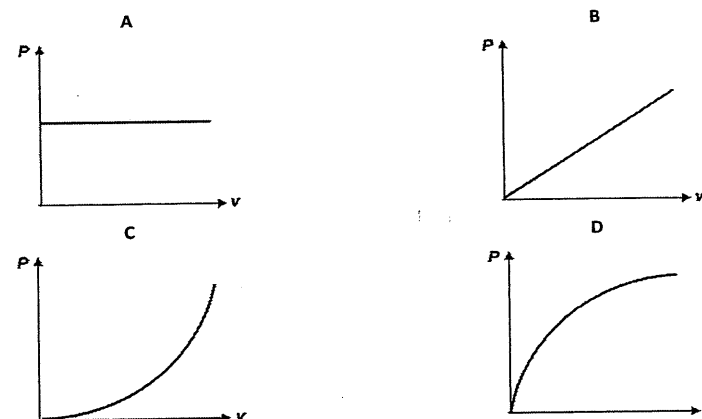
Which of the graphs correctly show how the kinetic energy (KE), elastic potential energy (EPE) and gravitational potential energy (GPE) would vary with time t ?



4. [ACJC/Prelims 2014/P1/12]

A car moving through air at velocity v experiences a resistive force F given by the expression $F = kv^2$ where k is a constant.

Which of the following graphs show how the power P supplied to the car will vary at various v to ensure that the car is moving without any acceleration?



5. [ACJC/Prelims 2014/P1/13]

A wind turbine has blades that sweep an area of 2000 m^2 . It converts the power available in the wind to electrical power with an efficiency of 50%.

What is the electrical power generated if the wind speed is 10 m s^{-1} ? (The density of air is 1.3 kg m^{-3} .)

- A 130 kW B 650 kW C 1300 kW D 2600 kW

6. [AJC/Prelims 2014/P1/9]

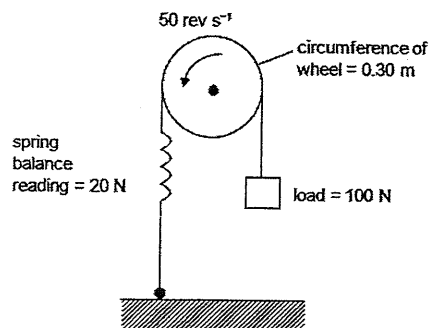
A man lifts a 10 kg sack of rice from a ground to above his head using his both hands. The sack of rice does not experience a change in kinetic energy between the two positions.

Which statement is correct?

- A The work done on rice sack by gravity is less than the work done on rice sack by man.
 B The work done on rice sack by gravity is more than the work done on rice sack by man.
 C The work done on rice sack by gravity is equal to the work done on rice sack by man.
 D There is no work done on the rice sack.

7. [CJC/Prelims 2014/P1/10]

The figure shows a wheel which is driven by an electric motor. A rope is fastened at one end to a spring balance. The rope passes over the wheel and supports a freely hanging load. When the wheel is turning anticlockwise at a steady speed, the balance reading is constant.

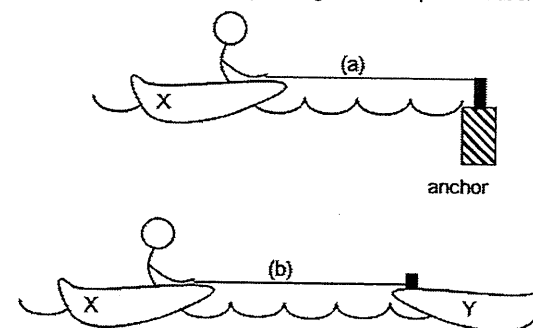


What is the output power of the motor?

- A 0.30 kW B 1.2 kW C 1.5 kW D 1.8 kW

8. [DHS/Prelims 2014/P1/10]

A boy on a boat X pulls on a rope with a constant force F over a duration of time t . The other end of the rope is either tied to an anchor on (a) the pier or (b) a freely floating boat Y of equal mass as shown.



W_a and W_b are the total work done by the boy during the time t whereas P_a and P_b are the average power output by the boy for case (a) and (b) respectively.

Which of the following is correct?

- A $W_a > W_b$ and $P_a > P_b$
 B $W_a = W_b$ and $P_a = P_b$
 C $W_a < W_b$ and $P_a < P_b$
 D $W_a > W_b$ and $P_a = P_b$

9. [HCI/Prelims 2014/P1/10]

A car of mass m has an engine that exerts a force of F on it. In a time t , the car travels a distance s and its speed increases from u to v .

What is the useful output power of the engine?

- A $\frac{Fs}{t}$ B $F(v - u)$ C Ft D $\frac{m(v^2 - u^2)}{2t}$

10. [MJC/Prelims 2014/P1/10]

The force resisting the motion of a car is taken as being proportional to the square of the car's speed. At a speed of 30 m s^{-1} , the resistive force is 1800 N.

What is the effective power required from the car's engine to maintain a steady speed of 15 m s^{-1} ?

- A 0.45 kW B 6.8 kW C 7.2 kW D 14 kW

11. [NJC/Prelims 2014/P1/11]

A space shuttle of mass m re-enters the Earth's atmosphere at an angle θ to the horizontal. Air resistance acting on the space shuttle causes it to travel at constant speed v . The heat-shield of the shuttle dissipates heat at a rate of P , so that the mean temperature of the shuttle remains constant.

Taking g as the relevant value of acceleration due to free fall, which expression is equal to P ?

- A mgv B $mgv \sin \theta$ C $\frac{1}{2}mv^2$ D $\frac{1}{2}mv^2 \sin^2 \theta$

12. [PJC/Prelims 2014/P1/10]

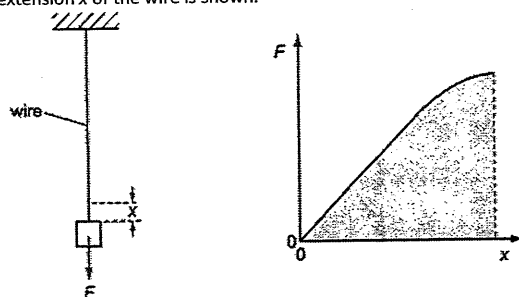
A boat moving at constant speed v through still water experiences a total frictional drag F .

What is the power developed by the boat?

- A $\frac{1}{2}Fv$ B Fv C $\frac{1}{2}Fv^2$ D Fv^2

13. [SAJC/Prelims 2014/P1/9]

A wire, fixed at its upper end, is subjected to an increasing load F by increasing the mass attached to its lower end. A graph of F against the extension x of the wire is shown.



The wire is stretched beyond its elastic limit.

What does the shaded area on the graph represent?

- A the amount of elastic potential energy stored in the wire
 B the amount of heat produced in the wire
 C the loss of gravitational potential energy of the mass
 D the work done by F on the wire

14. [TPJC/Prelims 2014/P1/9]

A speed-boat with three engines, each of power output 24 kW, can travel at a maximum speed of 12 m s^{-1} . The total drag D on the boat is related to the speed v of the boat by the relation shown.

$$D \propto v^2$$

What is the maximum speed of the boat when only two engines are working?

- A 8.0 m s^{-1} B 9.5 m s^{-1} C 9.8 m s^{-1} D 10.5 m s^{-1}

15. [VIC/Prelims 2014/P1/10]

An object of mass m is pushed against a spring, compressing it by a horizontal distance of d . The block is then released by launching it along a smooth horizontal surface, attaining a final speed of v . Another object of similar size and shape, but of mass $3m$, is pushed against the same spring and released in the same way.

What is the distance that the spring is compressed by the second object if its final speed is $3v$?

- A $1.7d$ B $5.2d$ C $9d$ D $15.6d$

16. {

- A
 B
 C
 D

17. {

A

B

C

D

18. {

A

B

C

D

19. {

A

B

C

D

20. {

A		
B		
C		
D		

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II. Structured Questions

1. [UC/Prelims 2014/P2/1]

Fig. 1.1 shows a section of a roller coaster track, not drawn to scale.

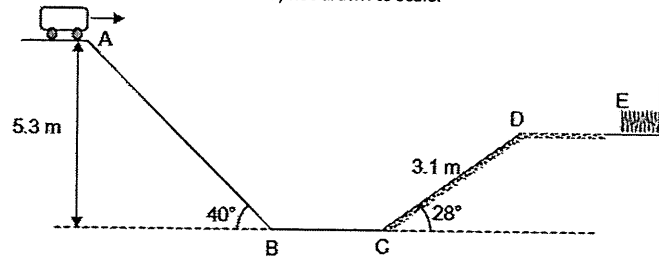


Fig. 1.1

The roller coaster enters the section of track at A, slides down a frictionless slope that makes an angle of 40° with the horizontal. It then travels along a straight, frictionless track BC before travelling along a rough incline CD. The incline CD has a length of 3.1 m and it makes an angle of 28° with the horizontal. It finally enters a rough braking zone DE where it is slowed down by the rough track. If the roller coaster still has some speed upon reaching E, a spring will eventually bring it to a complete stop.

At $t = 0$ s, a roller coaster of total mass 250 kg approaches A with a speed of 2.5 m s^{-1} .

- a. Determine the acceleration of the roller coaster along the slope AB. [2]
- b. Show that the speed of the roller coaster when it reaches B is 10.5 m s^{-1} . [2]
- c. Given that the roller coaster reaches C at $t = 2.5$ s, determine the length of the track between B and C. [3]
- d. Given that a constant frictional force of 290 N acts on the roller coaster along CD, determine the speed of the roller coaster when it reaches D. [4]
- e. After slowing down along the rough braking zone DE, the speed of the roller coaster was further reduced from the speed in {d.} to 3.0 m s^{-1} . It is then brought to rest by a spring whose force-compression (F - x) graph is shown in Fig. 1.2.

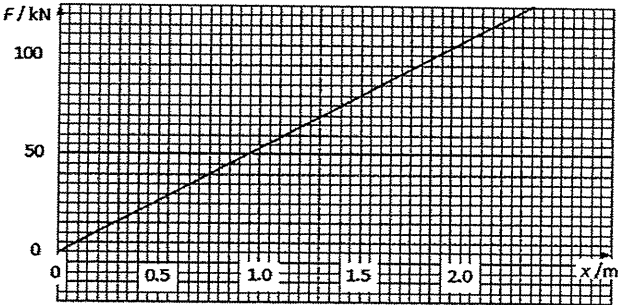


Fig. 1.2

Determine the maximum compression of the spring as it brings the roller coaster to a complete stop. [3]

[Total: 14]

2. [DHS/Prelims 2014/P2/1]

Trolley A of mass 400 g is moving at a constant speed of 2.5 m s^{-1} to the right as shown in Fig. 2.1.

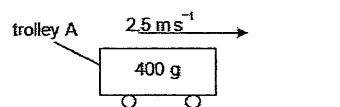


Fig. 2.1

- a. Show that the kinetic energy of trolley A is 1.3 J. [1]
- b. Trolley A hits a spring of force constant $k = 20 \text{ N m}^{-1}$ and compresses the spring from its equilibrium position before coming to rest momentarily.
 - i. Define elastic potential energy. [1]
 - ii. Calculate the maximum compression of the spring. [2]
 - iii. The length of the spring is then cut in half. Trolley A is, again, travelling with speed of 2.5 m s^{-1} and hits the spring. Suggest and explain the change, if any, on the maximum compression of the spring. [2]
- c. Two identical trolleys, A and B, start from rest at the same height above the ground and travel to the right on two frictionless tracks as shown in Fig. 2.2.

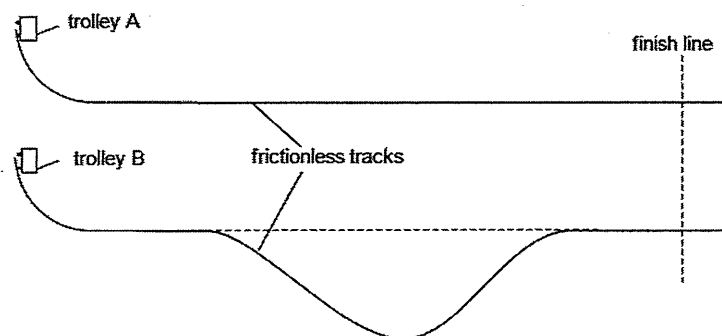


Fig. 2.2

State and explain which trolley will reach the finish line first.

[2]

[Total: 8]

3. [HCI/Prelims 2014/P3/1]*

Fig. 3.1 shows a toy gun with the spring, with spring constant $2.00 \times 10^3 \text{ N m}^{-1}$, in an unstretched state.

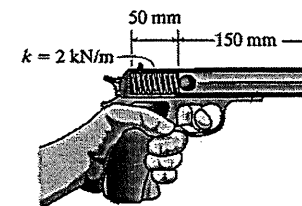


Fig. 3.1

As the trigger is pulled, the spring compresses by 2.00 cm.

- a. i. Show that the elastic potential energy that is stored in the spring is equal to 0.400 J. [1]
- ii. Instead of one spring, a gun is now designed where two springs of the same spring constant are put in series. The new spring system is again compressed by 2.00 cm. Explain whether the total elastic potential energy in the spring system is now more, less or equal to 0.400 J. [2]
- ab The spring from a. i is released and a 50.0 g pellet is shot away horizontally at a height of 1.50 m above the ground. The pellet slides through the barrel and experiences a friction of 0.100 N. [2]
- b. [2]
- i. Calculate the kinetic energy of the pellet when it leaves the barrel. [4]
- ii. Calculate the angle at which the pellet hits the ground. [3]
- cb As the pellet hits the ground, it takes 0.10 seconds for it to come to a full stop. [3]
- a. Calculate the average force exerted by the floor onto the pellet. [4]
- iii. [4]
- c. i. [4]

[Total: 811]

24. [SAJC/Prelims 2014/P3/7]

- a. A crate is dropped from a height onto a lorry. Explain how Newton's third law applies as the air resistance on the falling crate increases. [2]

The lorry then moves with the crate resting on the flat bed as shown in Fig. 4.1. A cat suddenly dashes across the road from a blind spot. On seeing the cat running across the road, the driver brakes and decelerates to rest.



Fig. 4.1

a. _____ [2]

b. _____ [2]

b. i. State what is meant by braking force. [14]

ii. A rough flat bed allows the crate to stay in the same position on the lorry when the lorry brakes.

On Fig. 7.1, draw an arrow to show the direction of the force that must act on the crate to allow this and label it F . [14]

c. State and explain two factors that affect the braking distance of the lorry. [2]

d. The lorry of mass 2000 kg has an engine which can deliver an average power of 170 kW.

Calculate the time in which the lorry can be accelerated from rest to a speed of 100 km h^{-1} on level ground. [2]

e. An automobile engine converts chemical energy released in the burning of the gasoline into mechanical energy that moves the engine piston and eventually the wheels. Only 25 % of the chemical energy released is converted to kinetic energy of the lorry.

Suggest why the lorry engine is roughly only about 25 % efficient. [1]

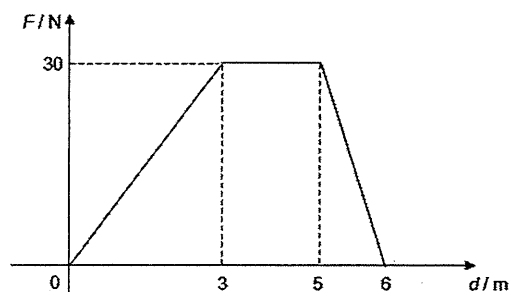
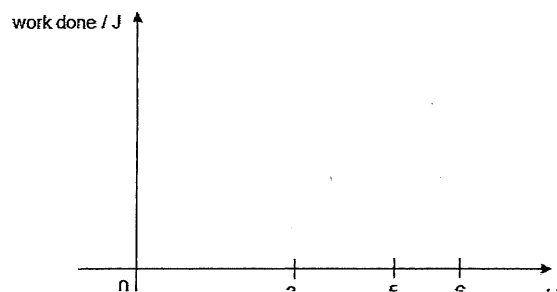
f. Fig. 4.2 shows the variation with displacement d of the force F applied to an object. F and d are in the same direction

Fig. 4.2

On Fig. 4.3, draw a graph showing the variation with d of the work done. Label the work done at the stated values of d . [4]

- g. An object A of mass m travelling with a speed u collides with another object B of mass $2m$ with a light spring of force constant k attached in front as shown in Fig. 4.4 below. The two bodies undergo an elastic collision during which the spring is compressed for a short duration. The bodies then separate after the collision.

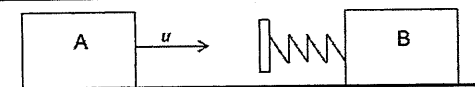


Fig. 4.4

If B is originally at rest on a smooth surface and no energy is lost to the surroundings.

- iii. Calculate the velocities of A and B after the collision, given that $u = 4.0 \text{ m s}^{-1}$. [4]

- ii. During the collision, the spring is compressed. By applying the principles of conservation of energy and conservation of linear momentum, show that the maximum compression (x_{max}) of the spring, is given by

$$x_{\text{max}} = u \sqrt{\frac{2m}{3k}}. \quad [3]$$

6. i. 1. [4]

[Total: 208]

10. i

- a. [2]

- b. [2]

- i. [4]

- ii. [4]

- iii. [4]

6. i. 1. [4]

[Total: 8]

III. Solutions and Answers

The suggested solutions and answers below are only for your reference and have not been verified to be error-free. If you have any doubt, please discuss with your tutor to clarify.

MCQs

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
D	C	C	C	B	C	B	C	D	B	B	B	D	D	B

- 1 D When the wire is stretched beyond its elastic limit, it will become slightly longer than its original length when it is unloaded i.e. the wire is permanently deformed. Hence the area under the curve will not be the amount of elastic potential energy stored in the wire. It is the work done by F on the wire.
- 2 C The area D_2 and D_3 represent the amount of energy that is needed to extend the elastic material.
- 3 C Recall the nature of KE and EPE graphs in SHM. GPE depends on reference point.
- 4 C Power = $Fv = kv^3$
- 5 B Power of wind = $\frac{1}{2} \rho A v^3 = \frac{1}{2} \times 1.3 \times 2000 \times 10^3 = 1.3 \text{ MW}$
Electrical power generated = $0.5 \times 1.3 = 650 \text{ KW}$
- 6 C Based on Energy Work Theorem and isolating rice sack as system,
Work done on rice sack by man + Work done on rice sack by gravity = Change in KE
Work done on rice sack by man = - Work done on rice sack by gravity
|Work done on rice sack by man| = |Work done on rice sack by gravity|
- 7 B Take the load and rope attached to load as one system, (assume massless rope)
To keep the load at a steady speed
tension due to spring balance + additional force due to wheel on the rope connected to load = mg
additional force due to wheel = $100 - 20 = 80 \text{ N}$
Velocity of rim of wheel = distance / time = circumference $\times 50 \text{ rev s}^{-1} = 0.30 \times 50 = 15 \text{ m s}^{-1}$
Output power = force $\times$ velocity = $80 \times 15 = 1200 \text{ W} = 1.2 \text{ kW}$

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- 8 C Since F is the same for both cases, both boat X will accelerate with the same magnitude. Hence for (b), both boat X and Y will move a distance of x compared to (a) where only boat X moves by x . Therefore, $W_a < W_b$ and $P_a < P_b$. The boy has to work doubly hard in (b).
- 9 D useful output power = rate of increase of KE = $\frac{\frac{1}{2}m(v^2 - u^2)}{t}$
- 10 B $F = \left(\frac{15}{30}\right)^2 1800$
 $P = Fv = \left(\frac{15}{30}\right)^2 1800 \times 15 = 6750 \text{ W}$
- 11 B $P = \frac{Fd}{t} = F_y v_y = mgv \sin \theta$
- 12 B At constant speed, the driving force on boat is equal to the total frictional drag F . $P = Fv$
- 13 D Area under force – extension graph is the work done by the force.
- 14 D At maximum speed, force by engine F = total drag D
 $P = Fv \Rightarrow P = kv^2 \times v \Rightarrow P = kv^3 \Rightarrow P \propto v^3$
 $\frac{3 \times 24}{2 \times 24} = \frac{12^3}{v^3} \Rightarrow v = 10.5 \text{ m s}^{-1}$
- 15 B By Conservation of Energy, Elastic PE = ΔKE
 $\frac{1}{2} kx^2 = \frac{1}{2} mv^2 \Rightarrow x \propto (\sqrt{m})v$ since k is the same.
 $\therefore \frac{x}{d} = \left(\sqrt{\frac{3m}{m}}\right) \frac{3v}{v} \Rightarrow x = (\sqrt{3})3d = 5.2d$

Structured Questions

1. [JC/Prelims 2014/P2/1]

a. $a = g \sin 40^\circ = 6.31 \text{ m s}^{-2}$

Alternative Solution

Net Force down the slope = $mg \sin 40^\circ$

Applying N2L: Net force = $ma \rightarrow mg \sin 40^\circ = ma$

$$a = g \sin 40^\circ = 6.31 \text{ m s}^{-2}$$

b. Kinematics method:

Distance travelled down the slope AB = $5.3 \div \sin 40^\circ = 8.245 \text{ m}$

Using $v^2 = u^2 + 2as$,

$$v_B^2 = 2.5^2 + 2(6.31)(8.245)$$

$$v_B = 10.5 \text{ m s}^{-1}$$

Alternative Solution

Using C.O.E., loss in GPE = Gain in KE

$$mg(5.3) = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$$

$$9.81 \times 5.3 = \frac{1}{2}v^2 - \frac{1}{2}(2.5)^2$$

$$v = 10.5 \text{ m s}^{-1}$$

c. Find time taken to travel down the slope AB, t_{AB} :

Using $v = u + at$,

$$10.5 = 2.5 + 6.30 t_{AB}$$

$$t_{AB} = 1.27 \text{ s}$$

For the constant speed motion along BC,

distance = speed $\times$ time:

$$\text{Length of BC} = 10.5 \times (2.5 - 1.27) = 12.9 \text{ m}$$

d. Work done by friction = frictional force $\times$ length of CD = $290 \times 3.1 = 899 \text{ J}$

By the Principle of Conservation of Energy for CD:

Loss in KE = Gain in GPE + WD against friction

$$\frac{1}{2}(250)(10.5^2 - v_D^2) = (250)(9.81)(3.1 \sin 28^\circ) + 899$$

$$v_D = 8.63 \text{ m s}^{-1}$$

Alternative Solution

Net force down the slope = $mg \sin 28^\circ + 290 = 1441 \text{ N}$

Applying $F = ma$: $1151 = ma \rightarrow a = 5.765 \text{ m s}^{-2}$

Applying $v^2 = u^2 + 2as$, taking up the slope as positive,

$$v^2 = 10.5^2 + 2(-5.765)(3.1)$$

$$v = 8.63 \text{ m s}^{-1}$$

e. Spring constant $k = F/x = 80000/1.5 = 53333 \text{ N m}^{-1}$

By the Principle of Conservation of Energy:

Loss in KE = Gain in EPE

$$\frac{1}{2}(250)3.0^2 = \frac{1}{2}(53333)(x)^2$$

$$x = 0.205 \text{ m}$$

2. [DHS/Prelims 2014/P2/1]

a. $KE = \frac{1}{2}mv^2 = \frac{1}{2}(0.4)(2.5)^2 = 1.25 = 1.3 \text{ J}$

b. i. See definitions in lecture notes

ii. loss in KE = gain in EPE

$$1.3 = \frac{1}{2}(20)(x^2)$$

$$x = 0.36 \text{ m}$$

iii. spring constant increases (by 2 times)

maximum compression reduces (is now 0.25 m)

c. horizontal distance is the same for both tracks

average horizontal speed of trolley B is higher than that of A,

trolley B first as (horizontal) speed of B is faster down the valley

3. [HCI/Prelims 2014/P3/1]*

a. i. $EPE = \frac{1}{2}kx^2 = 0.5 \times 2000 \times 0.0200^2 = 0.400 \text{ J}$

where EPE is the elastic potential energy, k is the spring constant and x is the extension (or, in this case, the compression) of the spring.

ii. For two equal springs in series, the new spring constant is half that of the individual springs.

Hence, less force is required to compress the spring and the potential energy in the spring system is now less than 0.400 J.

OR

If the total compression is 2.0 cm, the compression per spring is 1.00 cm.

The total energy is then given as $2 \times (0.5 \times 2000 \times 0.0100^2) = 0.200 \text{ J}$, which is less than 0.400 J.

b. i. By the principle of conservation of energy,

Elastic potential energy is converted into kinetic energy and heat (by friction)

$$KE = EPE - WD = 0.400 - (0.100 \times 0.170) = 0.400 - 0.017 = 0.383 \text{ J}$$

where KE is the kinetic energy of the pellet as it leaves the gun, EPE is the elastic potential energy as the spring is fully compressed and WD is the work down against friction.

ii. Initial KE = $\frac{1}{2}mv_k^2 = 0.383 \text{ J}$.

$$\text{Horizontal speed} = (2 \times 0.383 / 0.0500)^{1/2} = 3.914 \text{ m s}^{-1}$$

For the vertical motion, using $v_y^2 = u_y^2 + 2as$,

$$v_y = (2 \times 9.81 \times 1.50)^{1/2} = 5.425 \text{ m s}^{-1}$$

Since $\tan \theta = v_y / v_x$,

$$\theta = \tan^{-1}(5.425 / 3.914) = 54.2^\circ \text{ below the horizontal}$$

- c. Speed at the moment of impact is given by $v^2 = v_x^2 + v_y^2$

$$v = (3.914^2 + 5.425^2)^{1/2} = 6.690 \text{ m s}^{-1}$$

The change in momentum is $\Delta p = m \Delta v = 0.0500 \times (6.690 - 0) = 0.334 \text{ kg m s}^{-1}$

Since $F_{\text{avg}} = \Delta p / \Delta t$, $F_{\text{avg}} = 0.3345 / 0.10 = 3.34 \text{ N}$

This is the resultant force, due to the contact force and the weight of the pellet. The component of the weight in the direction of the motion is given by

$$W_{\parallel} = 0.050 \times 9.81 \times \sin 54.2^\circ = 0.397 \text{ N.}$$

Hence, the force exerted by the ground on the pellet is $3.345 + 0.397 = 3.74 \text{ N}$

4. [SAJC/Prelims 2014/P3/7]

- a. Any two from:

pull of Earth on object is equal and opposite to the pull of object on Earth

there is a force on the air (molecules) downwards/due to object

this force is increasing

the force the air exerts on the object and the force the object exerts on the air remain equal and opposite

- b. i. Braking force is the force exerted on the wheels by the road.

ii. arrow to the left

- c. road surface / conditions, brake system,

tyre surface / tread, speed of car,

mass, gradient of road,

for each factor an explanation to include the effect on the acceleration / force (and hence the braking distance) has to be given

- d. Average Power = $\frac{\text{work done}}{\text{time}} = \frac{\text{increase in K.E}}{\text{time}}$

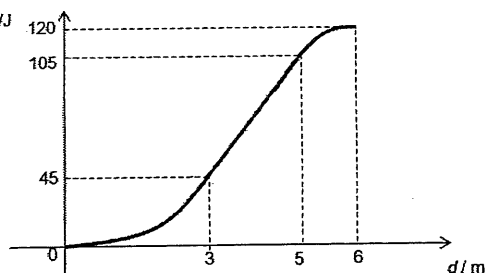
$$\text{time} = \frac{\text{increase in K.E}}{\text{average power}} = \frac{\frac{1}{2}(2000)(27.8)^2}{170 \times 10^3} = 4.54 \text{ s}$$

- e. Engine losses due to thermal energy loss

OR Engine losses due to friction between the engine parts

OR Engine losses due to coolant, exhaust gases

- f. work done / J



- g. i. $m_A u_A + 0 = m_A v_A + m_B v_B$

$$m(4.0) = m v_A + 2m v_B \quad \text{---(1)}$$

Elastic collision

$$u_A - 0 = v_B - v_A$$

$$4 = v_B - v_A \quad \text{---(2)}$$

$$\text{Solving, } v_A = -1.33 \text{ m s}^{-1}; \quad v_B = 2.67 \text{ m s}^{-1}$$

- ii. By principle of conservation of energy,

$$\frac{1}{2} m u^2 = \frac{1}{2} (m + 2m) v^2 + \frac{1}{2} k x_{\text{max}}^2$$

$$m u^2 = 3m v^2 + k x_{\text{max}}^2$$

By principle of conservation of momentum

$$m u = (m + 2m) v$$

$$v = \frac{u}{3} \quad \text{---(3)}$$

Sub (3) into (1)

$$m u^2 = 3m v^2 + k x_{\text{max}}^2 \Rightarrow m u^2 = 3m \left(\frac{u}{3}\right)^2 + k x_{\text{max}}^2 \Rightarrow x_{\text{max}} = u \sqrt{\frac{2m}{3k}}$$

